



2 - Coding Golf Assesment

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Q1 **Bubbled Sorting**

25 marks

Write a C code for performing Bubbled Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int A[5] = {8, 2, 3, 5, 7};
5     int i, j, temp;
6
7     // Bubble Sort
8     for (i = 0; i < 5 - 1; i++) {
9         for (j = 0; j < 5 - i - 1; j++) {
10             if (A[j] > A[j + 1]) {
11                 temp = A[j];
12                 A[j] = A[j + 1];
13                 A[j + 1] = temp;
14             }
15         }
16     }
17
18     printf("Sorted array: ");
19     for (i = 0; i < 5; i++) {
20         printf("%d ", A[i]);
21     }
22
23     return 0;
24 }
```

Input

stdin input

Output

Sorted array: 2 3 5 7 8



Run



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2:44



VoLTE

5G

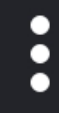
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Q2 Insertion Soring

25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int A[5] = {8, 2, 3, 5, 7};
5     int i, j, key;
6
7     for (i = 1; i < 5; i++) {
8         key = A[i];
9         j = i - 1;
10
11         while (j >= 0 && A[j] > key) {
12             A[j + 1] = A[j];
13             j--;
14         }
15
16         A[j + 1] = key;
17     }
18
19     printf("Sorted array: ");
20     for (i = 0; i < 5; i++) {
21         printf("%d ", A[i]);
22     }
23
24     return 0;
25 }
```

Input

stdin input

Output

Sorted array: 2 3 5 7 8

Run

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Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - n=7 A[] = {8, 2, 3, 5, 7,1,6}.

Your Code

```
1 #include <stdio.h>
2
3 void quickSort(int a[], int low, int high)
4 {
5     int i, j, pivot, temp;
6
7     if (low < high)
8     {
9         pivot = a[high];
10        i = low - 1;
11
12        for (j = low; j < high; j++)
13        {
14            if (a[j] < pivot)
15            {
16                i++;
17                temp = a[i];
18                a[i] = a[j];
19                a[j] = temp;
20            }
21        }
22
23        temp = a[i + 1];
24        a[i + 1] = a[high];
25        a[high] = temp;
26
27        int p = i + 1;
28
29        quickSort(a, low, p - 1);
30        quickSort(a, p + 1, high);
31    }
32 }
33
34 int main()
35 {
36     int a[] = {8, 2, 3, 5, 7, 1, 6};
37     int n = 7, i;
38
39     quickSort(a, 0, n - 1);
40
41     printf("Sorted array: ");
42     for (i = 0; i < n; i++)
43         printf("%d ", a[i]);
44
45     return 0;
46 }
```

Input

stdin input

Output

Sorted array: 1 2 3 5 6 7 8

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Q4 Merger Sort

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```
1 #include <stdio.h>
2
3 void merge(int a[], int low, int mid, int high)
4 {
5     int i = low, j = mid + 1, k = 0;
6     int temp[100];
7
8     while (i <= mid && j <= high)
9     {
10         if (a[i] < a[j])
11             temp[k++] = a[i++];
12         else
13             temp[k++] = a[j++];
14     }
15
16     while (i <= mid)
17         temp[k++] = a[i++];
18
19     while (j <= high)
20         temp[k++] = a[j++];
21
22     for (i = low, k = 0; i <= high; i++, k++)
23         a[i] = temp[k];
24 }
25
26 void mergeSort(int a[], int low, int high)
27 {
28     if (low < high)
29     {
30         int mid = (low + high) / 2;
31
32         mergeSort(a, low, mid);
33         mergeSort(a, mid + 1, high);
34
35         merge(a, low, mid, high);
36     }
37 }
38
39 int main()
40 {
41     int a[] = {8, 2, 3, 5, 7, 6, 4, 1};
42     int n = 8, i;
43
44     mergeSort(a, 0, n - 1);
45
46     printf("Sorted array: ");
47     for (i = 0; i < n; i++)
48         printf("%d ", a[i]);
49
50     return 0;
51 }
```

Input

stdin input

Output

Sorted array: 1 2 3 4 5 6 7 8